

PHYSICS 101 · WEEK 2 QUIZ

Physics 101 — Week 2

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. What did Newton's prism show about white light?

- ☐ A. White light is a single pure color that glass can tint into other colors.
- ☐ B. White light is made by mixing red and blue, the only two true colors.
- ☐ C. White light comes from the glass itself, which glows when sunlight hits it.
- ☐ D. White light is all the colors mixed together, and glass can spread them apart.

2. Why does a ripe tomato look red?

- ☐ A. It bends white light like a prism until only red is left.
- ☐ B. It makes its own red light, the way a light bulb glows.
- ☐ C. It absorbs the other colors and bounces the red back to your eye.
- ☐ D. It absorbs red light and reflects everything else.

3. What lies just past red in the rainbow, where your eyes can't see?

- ☐ A. Infrared — longer waves, felt as warmth.
- ☐ B. Violet — the last color the eye can see.
- ☐ C. X-rays — the waves that pass through bone.
- ☐ D. Ultraviolet — shorter waves, the sunburn part.

4. Why does the pulse oximeter shine TWO colors through a baby's foot?

- ☐ A. Loaded and empty hemoglobin absorb red and infrared differently; comparing them gives the percentage.
- ☐ B. Two colors are brighter than one, so enough light can make it through a baby's thicker foot.
- ☐ C. The red light warms the foot so that the infrared light can pass more easily through the skin.
- ☐ D. One color measures the heart rate while the other measures the temperature of the baby's foot.

5. What was Takuo Aoyagi's key idea in the 1970s?

- ☐ A. Measure the light with a camera instead of a detector, to see the color directly.
- ☐ B. Shine the light through the ear instead of the foot, because ears have no bone.
- ☐ C. Use a brighter red light, because babies' feet are thicker than adults' fingers.
- ☐ D. Use only the pulsing part of the light signal, because only artery blood pulses.

6. Why are bili lights blue?

- ☐ A. Blue light is the coolest color, so it doesn't overheat a baby lying under it for hours.
- ☐ B. Bilirubin absorbs blue light best, and the absorbed light changes it into a form the body can flush out.
- ☐ C. Blue is the opposite of yellow, and opposite colors cancel each other out on the skin.
- ☐ D. Blue light passes through the skin most easily and reaches the liver to switch it on.

7. What did Sister Jean Ward's babies in the sunny courtyard show?

- ☐ A. Babies in the sun gained weight faster than babies kept indoors.
- ☐ B. Sunlight raised the babies' oxygen levels as measured by the pulse ox.
- ☐ C. Skin the sun reached turned less yellow, while skin under the diaper stayed yellow.
- ☐ D. Fresh air stopped the babies' breathing pauses better than caffeine did.

8. Why do babies under bili lights wear eye shields?

- ☐ A. Because the shields reflect the light back onto the skin to speed up the cure.
- ☐ B. Because the bright light would wake them up and keep them from growing properly.
- ☐ C. As a precaution: a retina that's still growing doesn't need hours of bright blue light.
- ☐ D. Because the blue light would otherwise turn the whites of their eyes a yellow color.

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)